



Semantic Web Technology

Knowledge Base and Reasoning

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Outline

- The idea
- The technologies
- Resource Description Framework (RDF)
- RDF Schema (RDFS)



The idea (1/2)

- Sir Tim Berners-Lee – has originally coined the term “Semantic Web” and has inspired much research in this area.
- Important goals are indeed very similar to the goals of the Web in general: **to make knowledge widely accessible and to increase the utility of this knowledge by enabling advanced applications for searching, browsing, and evaluation**



- **Building models:** the quest for describing the world in abstract terms to allow for an easier understanding of a complex reality.
- **Computing with knowledge:** the endeavor of constructing reasoning machines that can draw meaningful conclusions from encoded knowledge.
- **Exchanging information:** the transmission of complex information resources among computers that allows us to distribute, interlink, and reconcile knowledge on a global scale.

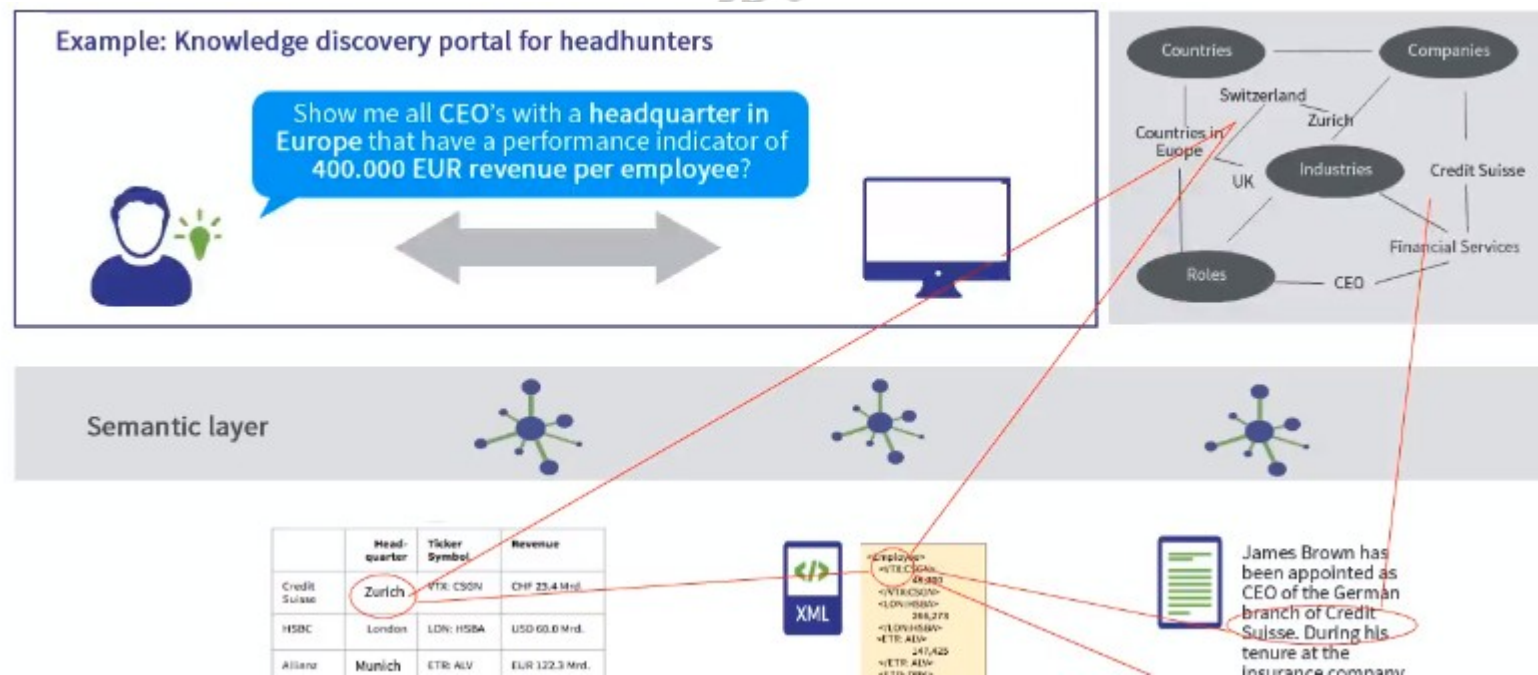


- The Semantic Web is an extension of the current Web in which information is given well-def (Berners-Lee et al. 2001)
- The Semantic Web provides a common framework that allows data to be shared and reused across application, enterprise, and community boundaries (W3C Semantic Web Activity Group)
- The Semantic Web is a collection of technologies and standards that allow machines to understand the meaning (semantics) of information on the Web.

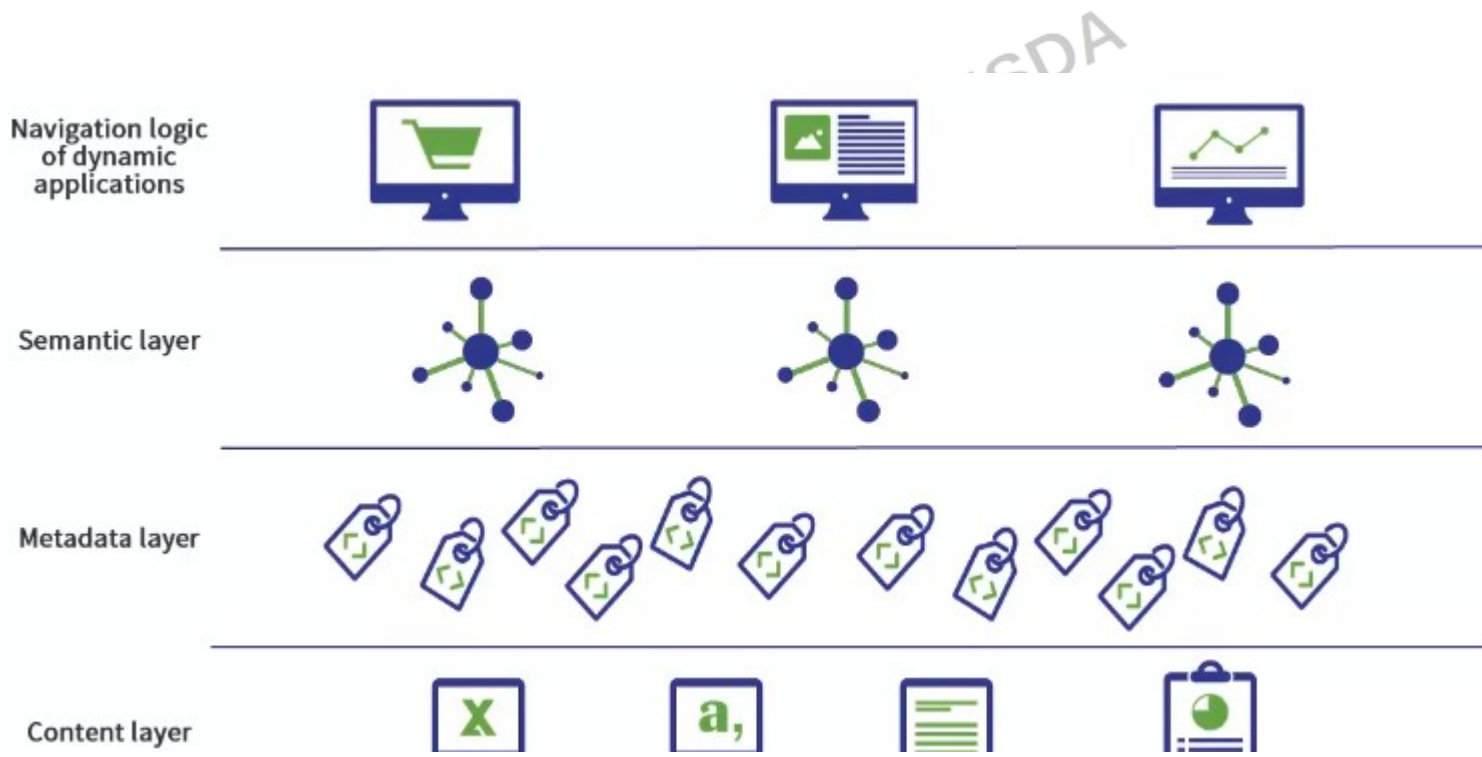


The idea (2/2)

- Data needs to be linked. Not in silos. Kind of integrated



Source: PoolParty



Source: PoolParty



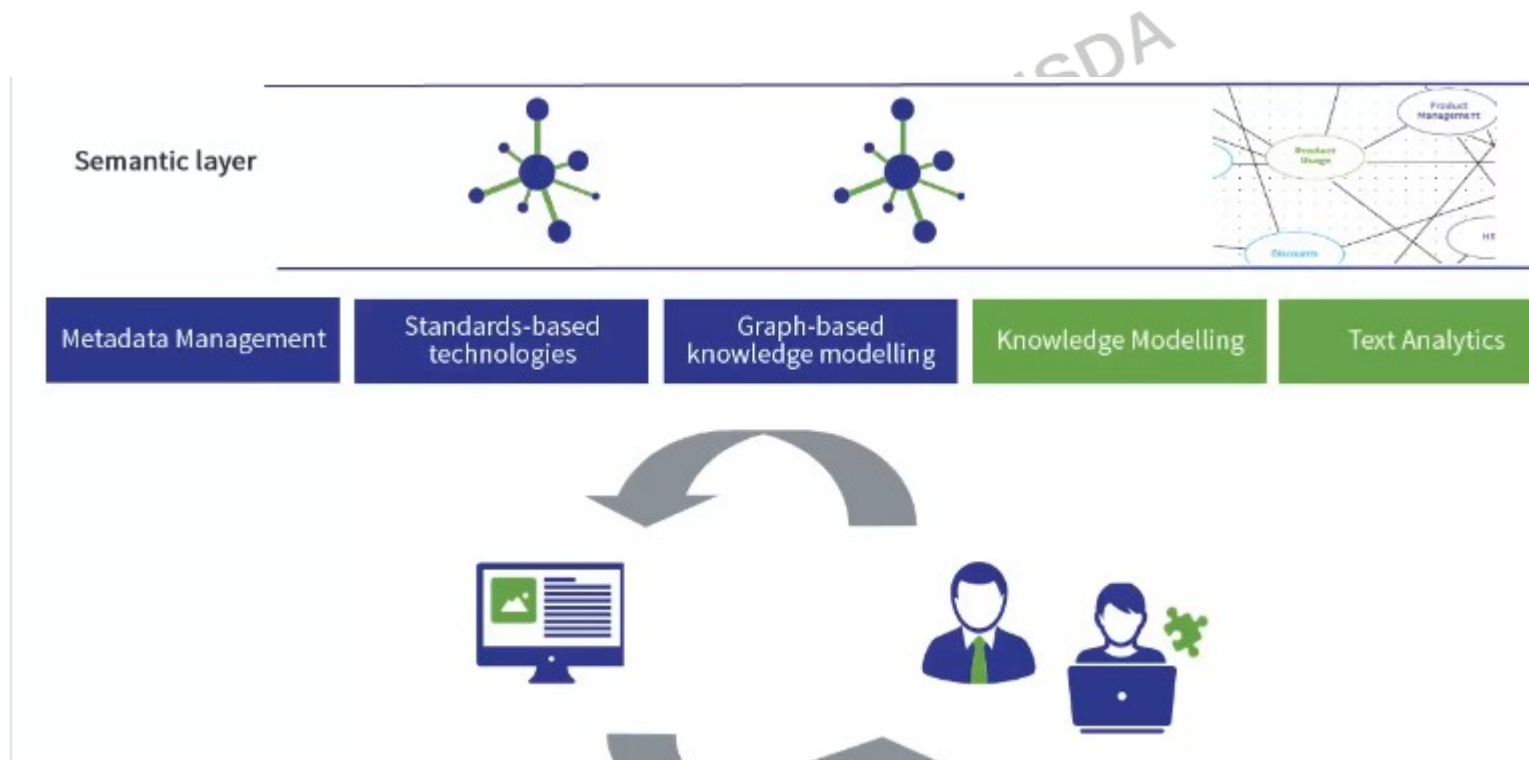
- Dublin Core Metadata → general
- Schema.org → SEO

Restricted to FATISDA



Descriptive metadata	Structural metadata	Administrative metadata	Rights Management metadata
Short product description The nex -generation Superfly Free expands in multiple diections thanks to a highly flexible midsole. Move your feet more dynamically with every stride. Brand: Superfly Free Size: ☐ S ☐ M ☐ L	Uniform Resource Identifier: http://myshop.com/sports/1245 Picture: png Text: html box	Created: 03-07-2015 Changed: 13-09-2016	Author: James Stuart Administrator: Anna Bell, Harry Golden, Susan Archer

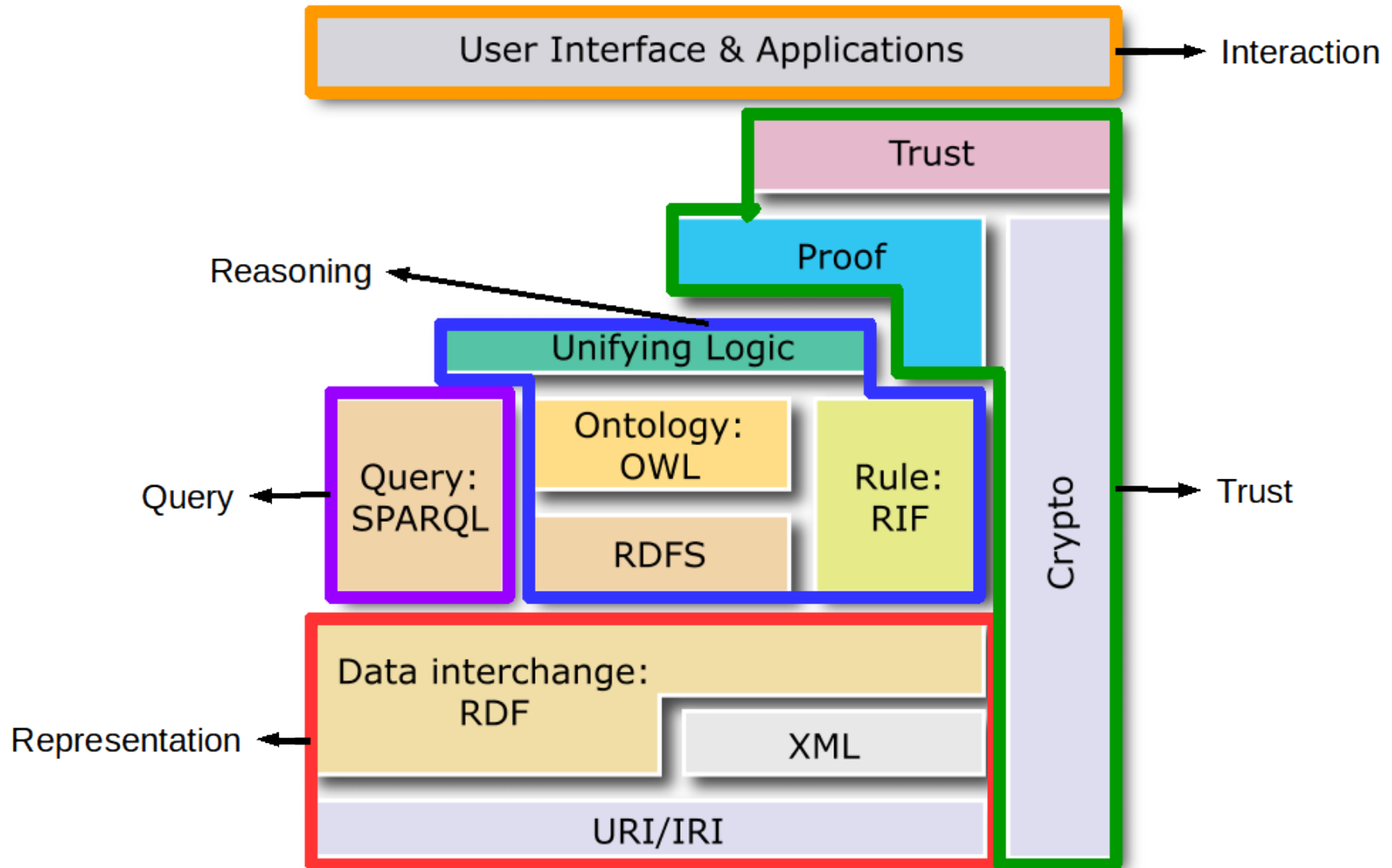
Source: PoolParty



Source: PoolParty



Outline





URI and XML

- The current technologies which are still used in the lower layer are the Internationalized **Resource Identifier (IRI)** and the **Extensible Markup Language (XML)**.
- IRI provides identification of each resource on the Web and **it guarantees that the resource is globally identified.**
- XML is a markup language **which provides a format for inserting metadata.**



- URLs can indeed be used as identifiers in RDF documents that talk about Web resources.
- In numerous other applications, however, the goal is not to exchange information about Web pages but about **many different kinds of objects**.
- scheme :[//authority]path [?query][#fragment]
- RDF allows for the encoding of data values which are not URIs, and it features so-called **blank nodes** which do not carry any name

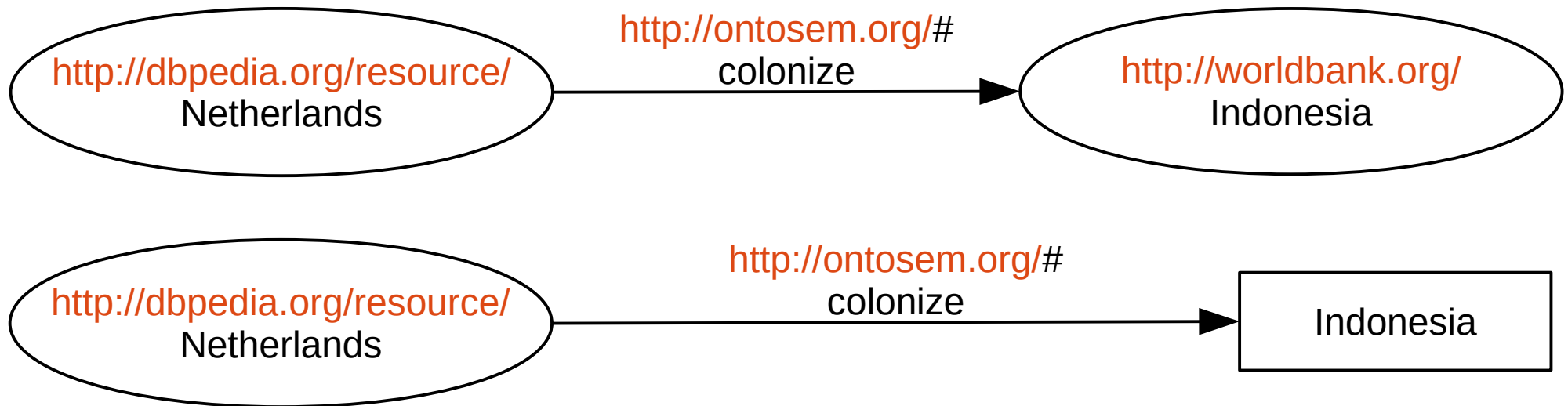


RDF

- RDF is the basic building block in the architecture of the Semantic Web.
- It is the standard data model in the Semantic Web, although some researchers name it as a language
- RDF describes the framework how to represent resources in the Semantic Web.
- It is a statement which consists of three elements Subject (S), Predicate (P) and Object (O)



- Each element can be formed either as **IRI, blank nodes, or data typed literals**. Although only objects can be formed as data typed literal
- **“Netherlands colonized Indonesia”**





RDF-Serialization

- RDF/XML
- RDF uses syntax of XML

```
<?xml version="1.0" encoding="UTF-8"?>
<rdf:RDF
  xmlns:osem="http://ontosem.org/#"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
  <rdf:Description rdf:about="http://dbpedia.org/resource/Netherlands">
    <osem:colonize rdf:resource="http://worldbank.org/Indonesia"/>
  </rdf:Description>
</rdf:RDF>
```



```
<?xml version="1.0" encoding="UTF-8"?>
<rdf:RDF
  xmlns:osem="http://ontosem.org/#"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
  <rdf:Description rdf:about="http://dbpedia.org/resource/Netherlands">
    <osem:colonize rdf:datatype="http://www.w3.org/2001/XMLSchema#literal">Indonesia
  </osem:colonize>
  </rdf:Description>
</rdf:RDF>
```



- **Turtle (.ttl).** It is used widely by the Semantic Web applications. Turtle allows RDF to be completely written in a compact and natural text form, with abbreviations for common usage patterns and datatypes.

```
@prefix dbpedia:<http://dbpedia.org/resource/> .  
@prefix wb:<http://worldbank.org/> .  
@prefix osem:<http://ontosem.org/#> .  
dbpedia:Netherlands osem:colonize wb:Indonesia .
```

```
@prefix dbpedia: <http://dbpedia.org/resource/> .  
@prefix osem: <http://ontosem.org/#> .  
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .  
dbpedia:Netherlands osem:colonize "Indonesia"^^xsd:literal .
```



- **N-Triples (.nt)**. It is a turtle format which does not use the abbreviations to simplify the IRI

<http://dbpedia.org/resource/Netherlands>

<http://ontosem.org/#colonize>

<http://worldbank.org/Indonesia> .

<http://dbpedia.org/resource/Netherlands>

<http://ontosem.org/#colonize>

"Indonesia"^^<http://www.w3.org/2001/XMLSchema#literal> .



- **RDFa** is the serialization of RDF in case we need to embed structured data into HTML pages directly

```
<div xmlns="http://www.w3.org/1999/xhtml"
  prefix="
    rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#
    osem: http://ontosem.org/#
    rdfs: http://www.w3.org/2000/01/rdf-schema#" >
  <font typeof="rdfs:Resource" about="http://dbpedia.org/resource/Netherlands">Netherlands
    <font rel="osem:colonize" resource="http://worldbank.org/Indonesia">colonize Indonesia</font>
  </font> </div>
```

```
<div xmlns="http://www.w3.org/1999/xhtml"
  prefix="
    rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#
    osem: http://ontosem.org/#
    xsd: http://www.w3.org/2001/XMLSchema#
    rdfs: http://www.w3.org/2000/01/rdf-schema#" >
  <font typeof="rdfs:Resource" about="http://dbpedia.org/resource/Netherlands">Netherlands
    <font property="osem:colonize" datatype="xsd:literal" content="Indonesia">colonize Indonesia</font>
  </font> </div>
```



- **N-quads (.nq)**. It is an RDF, which uses a fourth element. The added element is a triple context. Therefore, N-Quads consists of subject, predicate, object and triple context
- Triple context can be formed as an IRI or a blank node

`<http://dbpedia.org/resource/Netherlands> <http://www.ontosem.org/#colonize>
<http://worldbank.org/Indonesia>
_:C1 .`

`<http://dbpedia.org/resource/Netherlands> <http://www.ontosem.org/#colonize>
"Indonesia"http://www.w3.org/2001/XMLSchema#literal
_:C1 .`



- **Named Graph (.trig).** It is an extension of the Turtle format, which has a graph name.
- It consists of a sequence of directives, triple statements, graph statements which contain triple-generating statements and an optional blank line.

```
@prefix dbpedia: <http://dbpedia.org/resource/> .
```

```
@prefix osem: <http://ontosem.org/#> .
```

```
@prefix wb: <http://worldbank.org/> .
```

```
GRAPH _:G1 { dbpedia:Netherlands osem:colonize wb:Indonesia }
```

```
@prefix dbpedia: <http://dbpedia.org/resource/> .
```

```
@prefix osem: <http://ontosem.org/#> .
```

```
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
```

```
GRAPH _:G1 { dbpedia:Netherlands osem:colonize "Indonesia"^^:xsd:literal }
```



- **JSON-LD (.json)**. It is an RDF, which completely written in a form compatible with the JavaScript Object Notation (JSON)

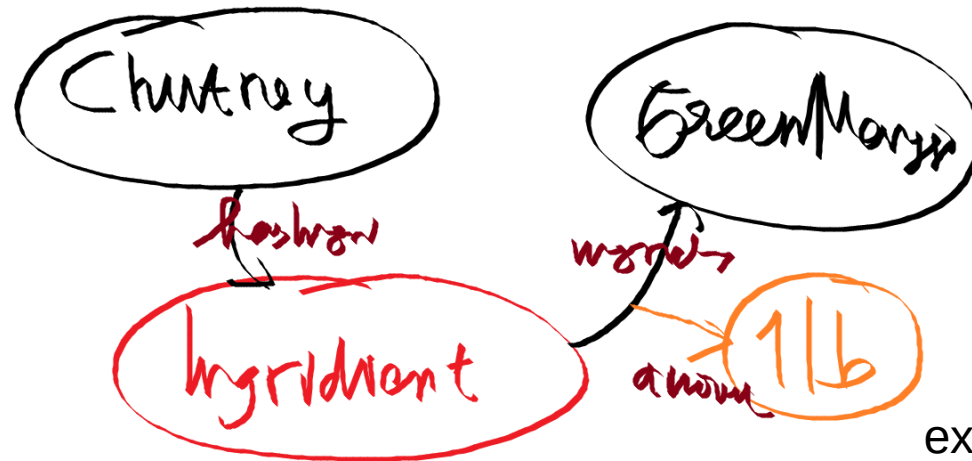
```
{
  "@context": {
    "dbpedia": "http://dbpedia.org/resource/",
    "osem": "http://ontosem.org/#",
    "rdf": "http://www.w3.org/1999/02/22-rdf-syntax-ns#",
    "rdfs": "http://www.w3.org/2000/01/rdf-schema#",
    "wb": "http://worldbank.org/",
    "xsd": "http://www.w3.org/2001/XMLSchema#"
  },
  "@id": "dbpedia:Netherlands",
  "osem:colonize": {
    "@id": "wb:Indonesia"
  }
}
```

```
{
  "@context": {
    "dbpedia": "http://dbpedia.org/resource/",
    "osem": "http://ontosem.org/#",
    "rdf": "http://www.w3.org/1999/02/22-rdf-syntax-ns#",
    "rdfs": "http://www.w3.org/2000/01/rdf-schema#",
    "xsd": "http://www.w3.org/2001/XMLSchema#"
  },
  "@id": "dbpedia:Netherlands",
  "osem:colonize": {
    "@type": "xsd:literal",
    "@value": "Indonesia"
  }
}
```

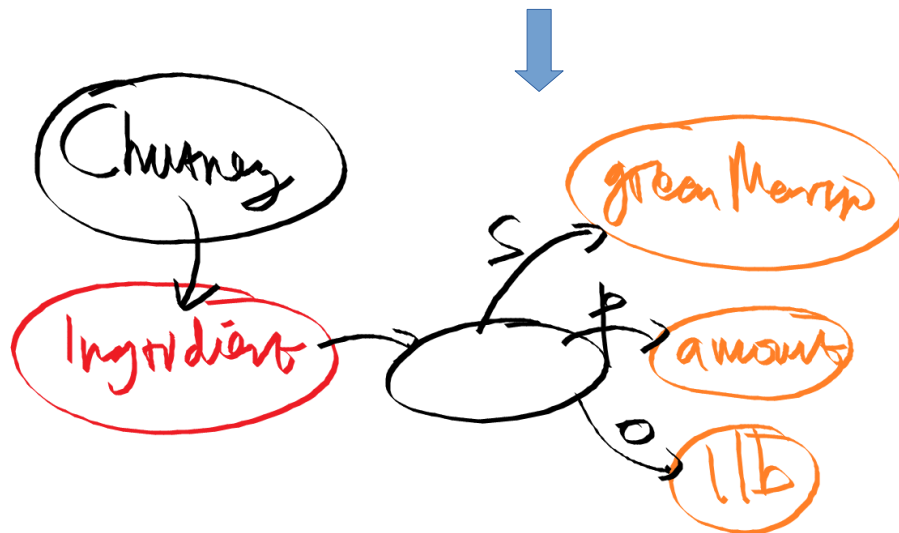


Reification

- The standard RDF, which is formed by three elements S, P, and O, **has a limitation in representing complex data.**
- Therefore, **Reification approach was introduced**, but it has a limitation regarding inefficiency issue.
- **Reification uses a surrogate term**, then uses the surrogate term to create reified statements.



```
ex:Chutney ex:Ingrident _:n .
_:n rdf:subject ex:greenMango .
_:n theory rdf:predicate ex:amount .
_:n theory rdf:object ex:1lb .
```





- “Netherlands colonized Indonesia for 350 years”

@prefix dbpedia:<http://dbpedia.org/resource/> .

@prefix wb:<http://worldbank.org/> .

@prefix osem:<http://ontosem.org/#> .

@prefix xsd:<http://...> .

_:n0 rdf:subject dbpedia:Netherlands .

_:n0 rdf:predicate osem:colonize .

_:n0 rdf:object wb:Indonesia .

_:n0 osem:duration “350”^^xsd:integer .



Datatype

- By now, **literals were untyped**, interpreted as **strings** (i.e.: "02"<"100"<"11"<"2")
- Typing literals with datatypes **allows for more adequate** (semantic = meaning-appropriate) treatment of values datatypes denoted by URIs and can be freely chosen frequently: xsd datatypes from XML
- **Syntax of typed literal: "datavalue" ^^ datatype-URI**



- Ex: “Nana has surname Sunana, was born on 01.01.2010”
- @prefix xsd:
<http://www.w3.org/2001/XMLSchema#> .
@prefix ex: <http://www.ex.org/>
ex:Nana ex:surname “Sunana”^^xsd:string .
ex:Nana ex:birthDate “2010-01-01”^^xsd:date .



- Many – valued relationship

•



@prefix ex: <http://example.org/> .

@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .

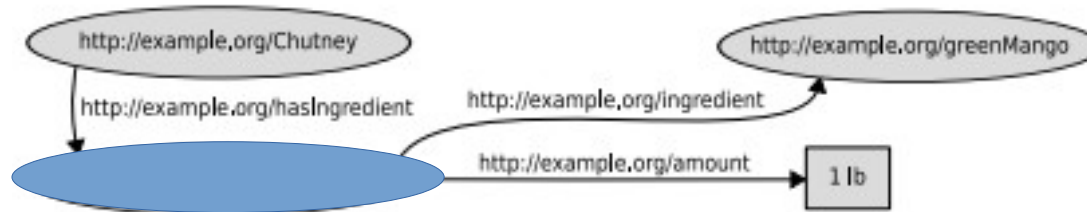
ex:Chutney ex:hasIngredient ex:ingredient1 .

ex:ingredient1 rdf:value ex:greenMango;

ex:amount "1lb" .



- Using blank node



```
@prefix ex: <http://example.org/> .  
ex:Chutney ex:hasIngredient _:id1 .  
_:id1 ex:ingredient ex:greenMango;  
ex:amount "1lb" .
```

or

```
@prefix ex: <http://example.org/> .  
ex:Chutney ex:hasIngredient  
[ ex:ingredient ex:greenMango; ex:amount "1lb" ] .
```

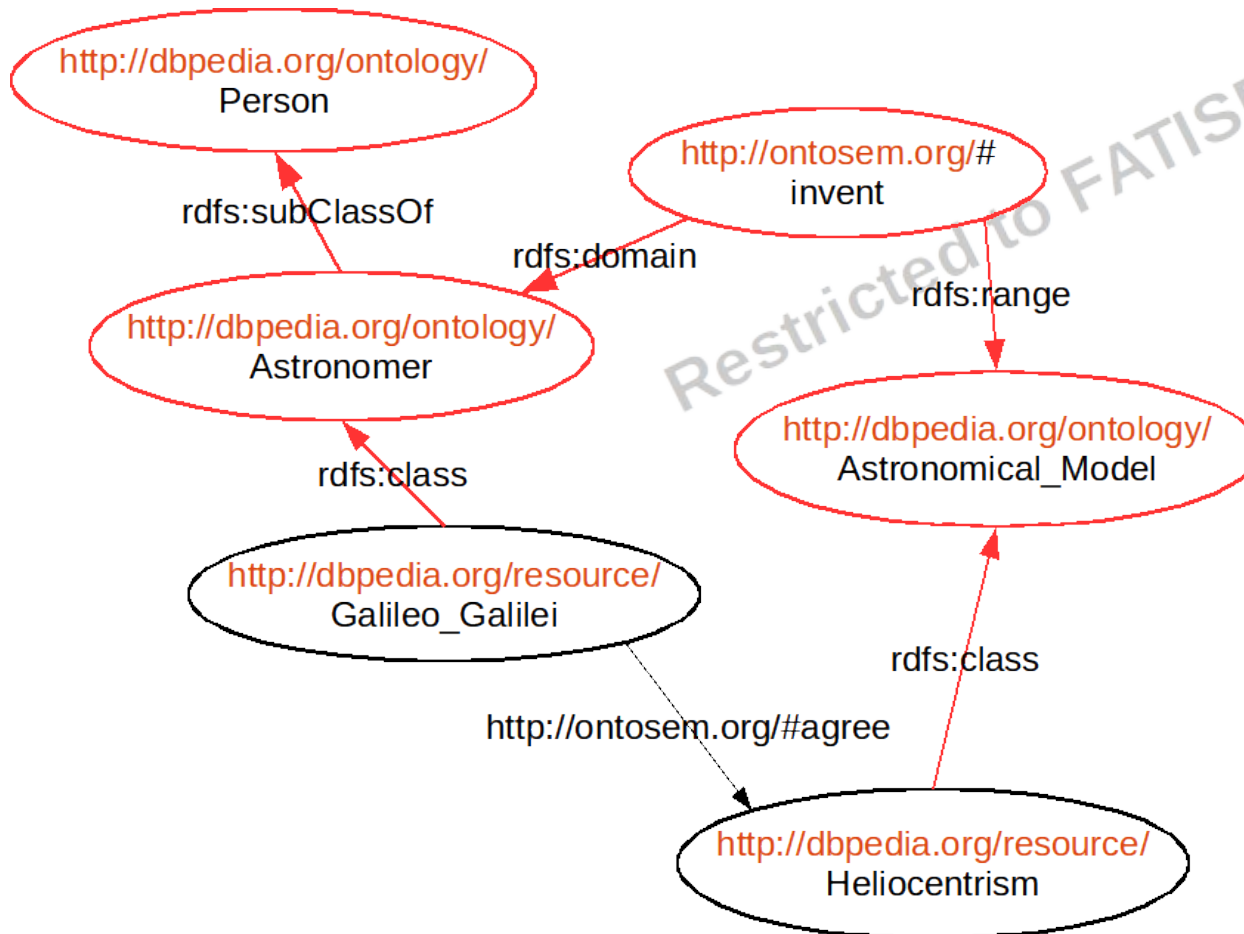


RDFS

- RDFS is an extension of RDF, **which provides vocabularies for RDF**
- **It adds a schema in RDF**. It describes resources and the relationship between resources.
- Vocabularies define **concepts and relationships which are used to describe and to represent an area of concern**. Users can define their vocabularies by using RDFS



- **Some core vocabularies** are `rdfs:Resource`, `rdfs:Class`, `rdfs:Literal`, `rdf:Property`, `rdf:Statement`, `rdfs:subClassOf` and `rdfs:subPropertyOf`.
- **The complete vocabularies are released as a part of the W3C recommendation**



The figure is the example of triple RDF which is extended by RDFS (red line). The resource “Galileo_Galilei” has type class “Astronomer”. The class “Astronomer” is a subclass of the class “Person”. The resource “Heliocentrism” has type class “Astronomical_Model”. The property “invent” has class domain “Astronomer” and has class range “Astronomical_Model”.



- We can see that RDFS adds deeper semantics than RDF.
- RDFS aims to define additional descriptions of relationships among resources. Thus, it is a set of vocabularies which provide a schema for RDF.



@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix res: <http://dbpedia.org/resource> .
@prefix ont: <http://dbpedia.org/ontology> .
@prefix osem: <http://ontosem/#> .

res:Galileo_Galilei osem:agree res:Heliocentrism .
res:Galileo_Galilei rdfs:class ont:Astronomer .
res: Heliocentrism rdfs:class ont:Astronomical_Model .
ont:Astronomer rdfs:subClassOf ont:Person .
osem:invent rdfs:domain ont:Astronomer .
osem:invent rdfs:range ont:Astronomical_Model .



Class and Instances

- Instances is related to individual (data) and class is related to the schema of individual
`wb:Indonesia rdf:type dbpedia:Country .`
- Instances is not exclusive only a member of one class
`wb:indonesia rdf:type dbpedia:G20 .`
- Class can be identified as a class-identifier
`dbpedia:Country rdf:type rdfs:Class .`



subClasses

- To express that one class is a part of the other class
- Use `rdfs:subClassOf`
`dbpedia:G20 rdfs:subClassOf dbpedia:Country .`
- It makes RDFS is more hierarchical
- `rdf:type` $\rightarrow \square \in \square$
- `rdfs:subClassOf` $\rightarrow \square \subseteq \square$



Property

- Technical term for **Relationship**
- Property names usually occur in predicate position in factoid RDF triples characterize, how two resources are related
- Mathematically: set of pairs

friendship = {(A,B),(B,C),...}

- URI can be marked as property name by typing it accordingly:

osem:friendship rdf:type rdfs:Property .



subPropertyOf

- In analogy to subclass of relationships representation in RDFS via `rdfs:subPropertyOf` e.g.:
`osem:roommate rdfs:subPropertyOf osem:friendship .`



Property Restrictions

- Properties may give hints what types the linked resources have, e.g. we know that `osem:colonize` connects a colonizing-country to a colonized-country
- i.e., for all URIs `a`, `b` where we know
`dbpedia:Netherlands osem:colonize wb:Indonesia` .
- We want to automatically follow:
`dbpedia:Netherlands rdf:type history:ColonizingCountry` .
`wb:Indonesia rdf:type history:ColonizedCountry` .
- This generic correspondence can be encoded in RDFS:
`osem:colonize rdfs:domain history:ColonizingCountry` .
`osem:colonize rdfs:range history:ColonizedCountry` .



- With property restrictions, **semantic interdependencies between properties and classes can be specified**
- Caution: property restrictions are interpreted globally and conjunctively, e.g.
ex:colonize rdfs:range ex:Island .
ex:colonize rdfs:range ex:Country .
- It means: everything which is colonized by a country is both an island and a country thus: always use most generic classes for domain/range statements



Additional Information

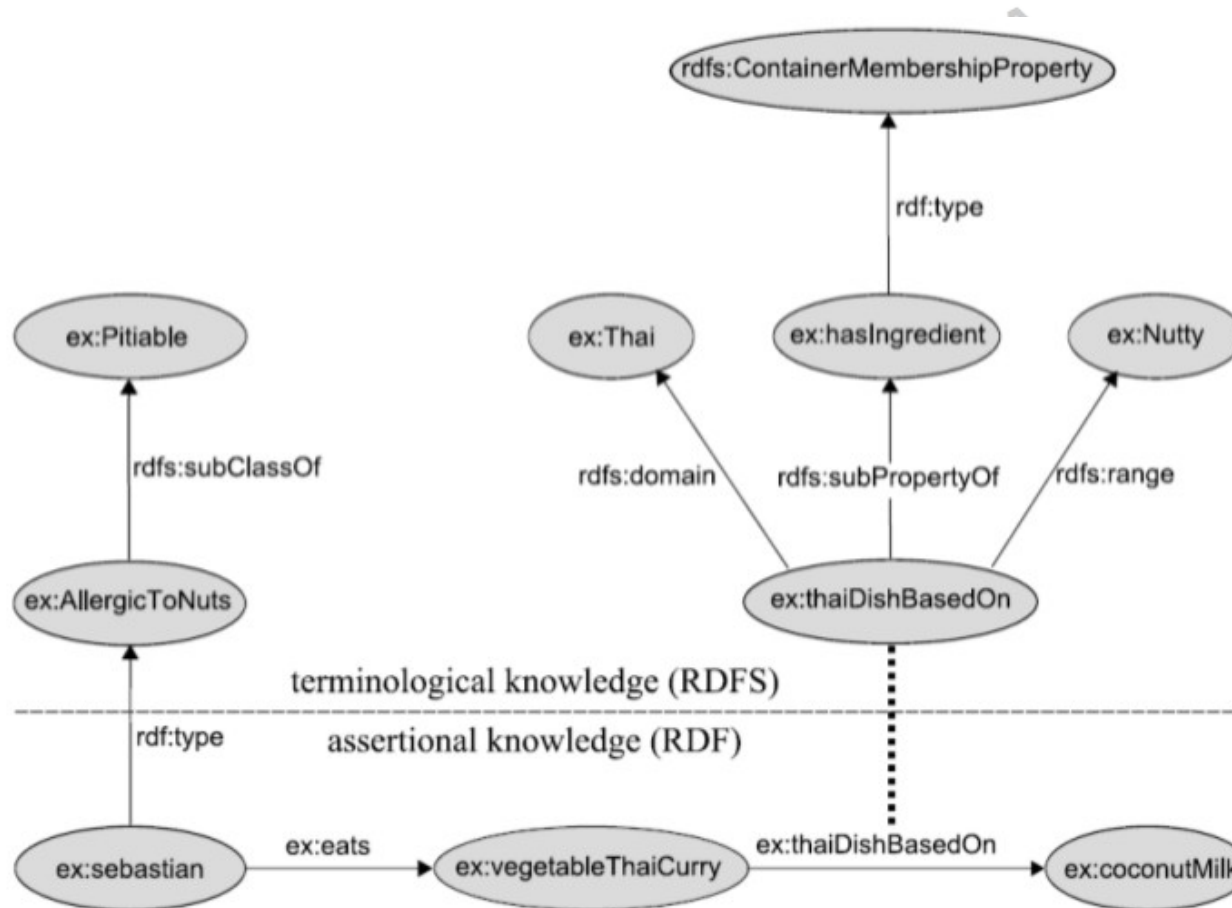
- Used to add human-readable information (comments or names) for compatibility reasons graph-based representation recommended; set of properties for that purpose:
- **rdfs:label** assigns an alternative name (encoded as literal) to an arbitrary resource
- **rdfs:comment** assigns a more comprehensive comment (also literal)
- **rdfs:seeAlso**, **rdfs:definedBy** refer to resources (URIs) containing further information about the subject resource



```
:
xmlns:wikipedia="http://en.wikipedia.org/wiki/"
:
<rdfs:Class rdf:about="&ex;Primates">
<rdfs:label xml:lang="en">primates</rdfs:label>
<rdfs:comment>
    Order of mammals. Primates are characterized by an
    advanced brain. They mostly populate the tropical
    earth regions. The term 'Primates' was coined by
    Carl von Linné.
</rdfs:comment>
<rdfs:seeAlso rdf:resource="&wikipedia;Primates" />
<rdfs:subClassOf rdf:resource="&ex;Mammalia" />
</rdfs:Class>
```



RDF vs RDFS

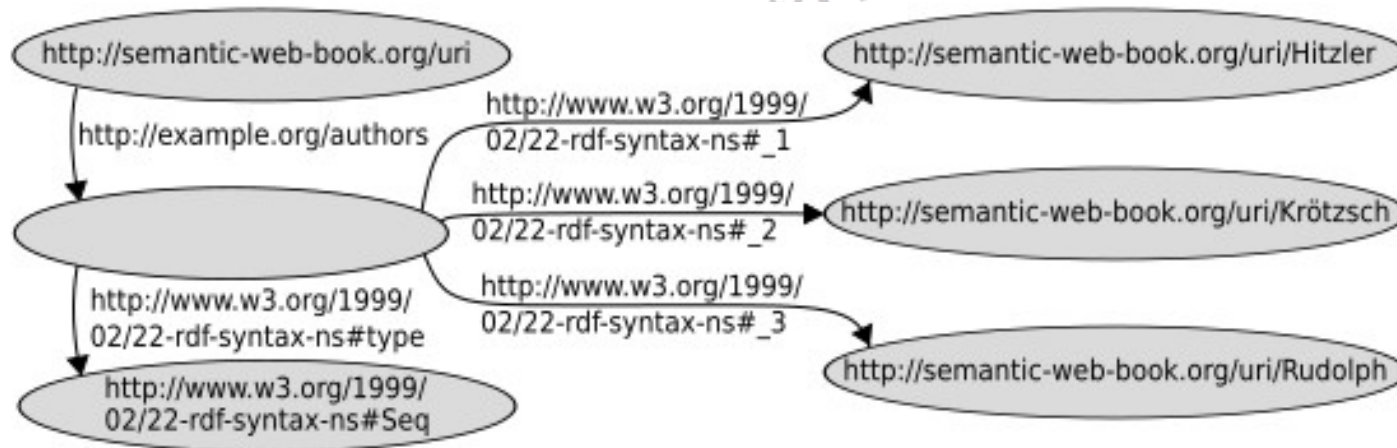




The other data structure

- Open Lists (Container)

-



By rdf:type we assign a list type to the root node

rdf:Seq – ordered list (sequence)

rdf:Bag – unordered list

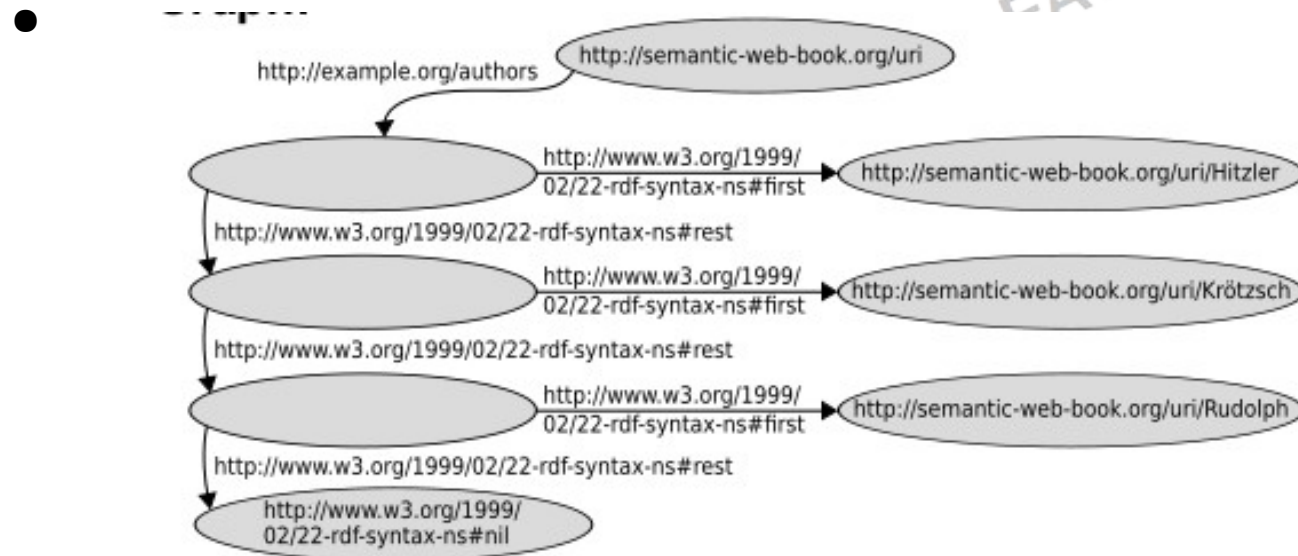
rdf:Alt – set of alternatives or choices



`<rdf:Description rdf:about="http://semantic-web-book/uri">
<ex:authors>
<rdf:Seq>
<rdf:li rdf:resource="http://semantic-web-book.org/uri/Hitzler" />
<rdf:li rdf:resource="http://semantic-web-book.org/uri/Krötzsch" />
<rdf:li rdf:resource="http://semantic-web-book.org/uri/Rudolph" />
</rdf:Seq>
</ex:authors>
</rdf:Description>`



- Closed Lists (Collections)



@prefix book: <http://semantic-web-book.org/> .
book:uri <http://example.org/authors>
(book:uri/Hitzler book:uri/Krötzsch book:uri/Rudolph) .



<rdf:Description rdf:about="http://semantic-web-book/uri">
<ex:authors rdf:parseType="Collection">
<rdf:Description
rdf:about="http://semantic-web-book.org/uri/Hitzler" />
<rdf:Description
rdf:about="http://semantic-web-book.org/uri/Krötzsch" />
<rdf:Description
rdf:about="http://semantic-web-book.org/uri/Rudolph" />
</ex:authors>
</rdf:Description>



Further Reading

- A Semantic Web Primer. The MIT Press Cambridge, Massachusetts, London, England. Grigoris Antoniou and Frank van Harmelen.
- <https://www.w3.org/TR/rdf-primer/>
- <https://www.w3.org/TR/rdf-schema/>